

CardioEquation — System Architecture

Patient-Specific ECG Generation using Diffusion Transformers

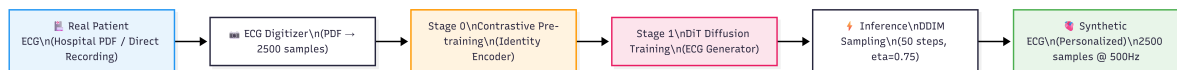
A comprehensive technical overview for academic review.

1. What Is CardioEquation?

CardioEquation is a **deep learning system that generates personalized, realistic ECG signals** for a specific patient. Given a short segment of a real patient's ECG as input, the system generates a new synthetic ECG that:

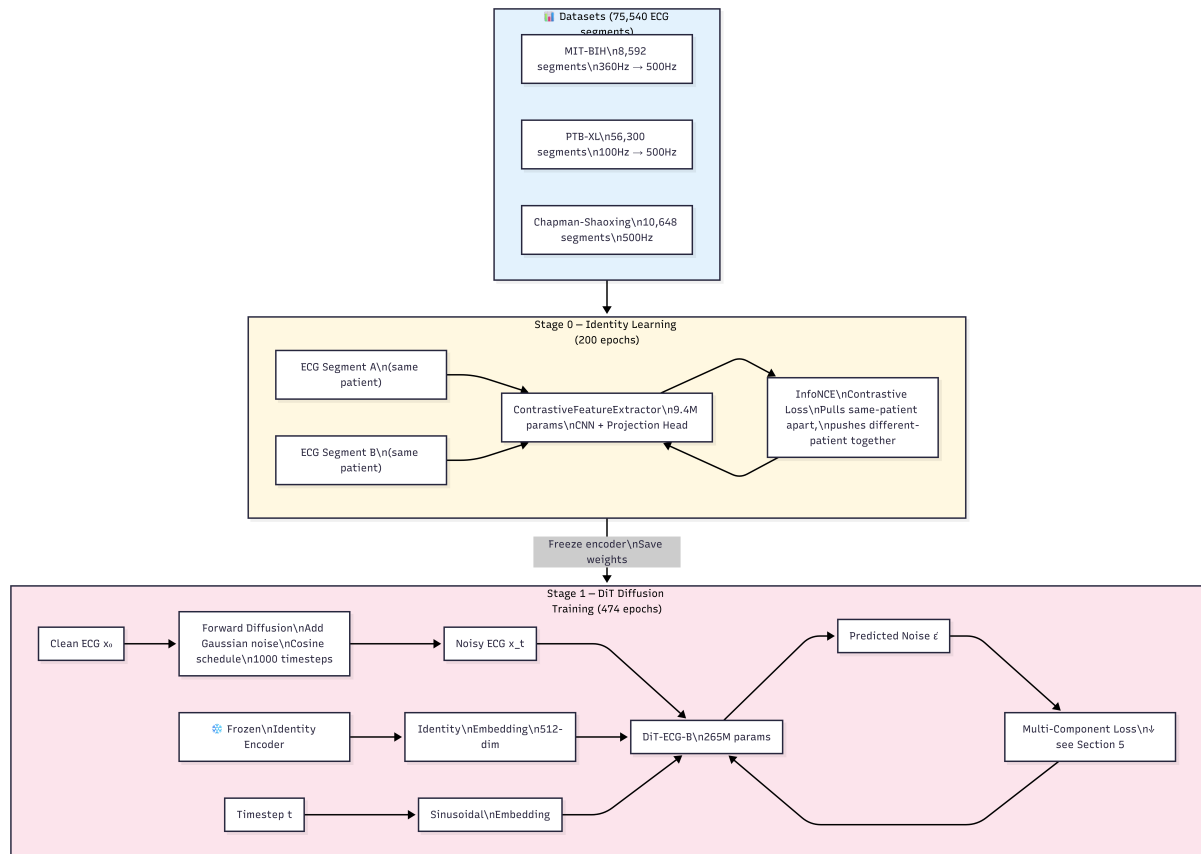
- Looks like **that patient's heart** (preserving morphology — P-wave shape, QRS complex, T-wave)
- Has **realistic rhythm** (heart rate, HRV)
- Is **clinically useful** for training diagnostic AI models, data augmentation, and privacy-preserving medical data sharing

2. High-Level System Overview



3. The Two-Stage Training Pipeline

Overview



4. Core Model: DiT-ECG-B Architecture

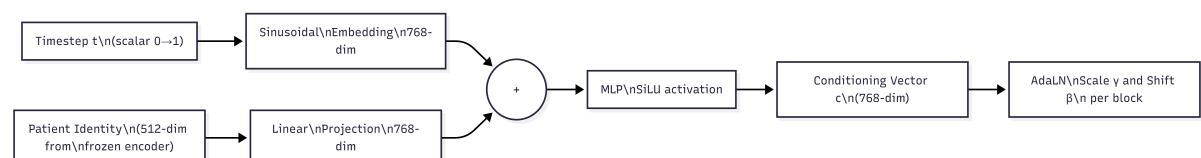
The generator is a **Diffusion Transformer** adapted from DiT (Peebles & Xie, 2022) for 1D ECG signals.

4.1 Input Representation



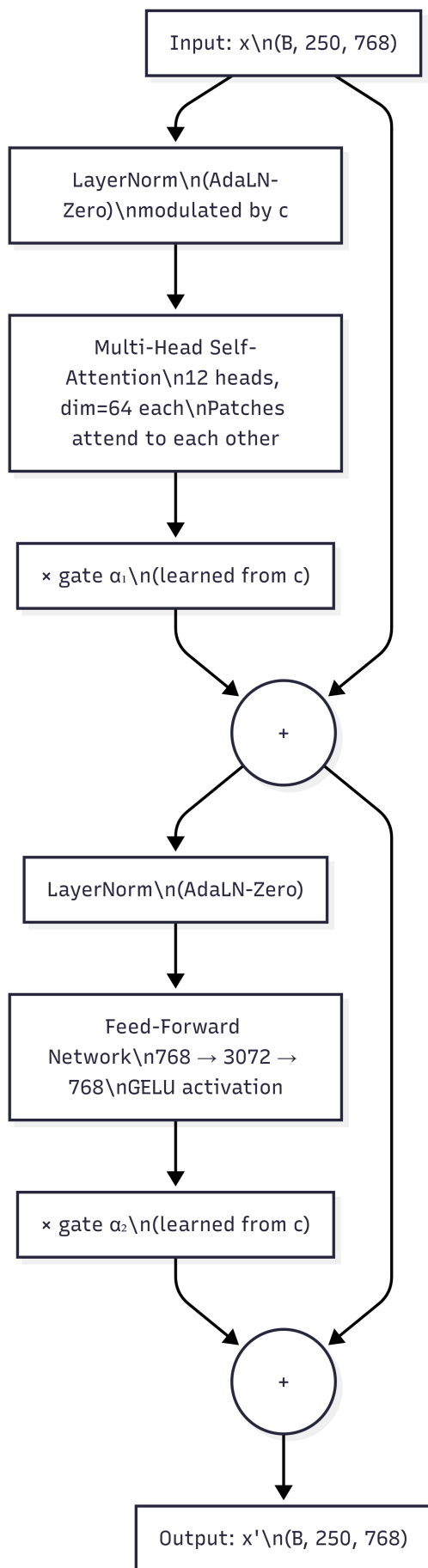
Why patches? Processing 2500 raw samples through self-attention would require $2500^2 = 6.25\text{M}$ attention pairs per layer. With patch size 10, this drops to $250^2 = 62,500$ — **100× reduction**.

4.2 Conditioning Mechanism (AdaLN-Zero)



AdaLN-Zero: Instead of fixed LayerNorm, the model learns to modulate each layer differently based on the conditioning vector. Starts at zero (no modulation) and learns the right amount during training.

4.3 Single DiT Block



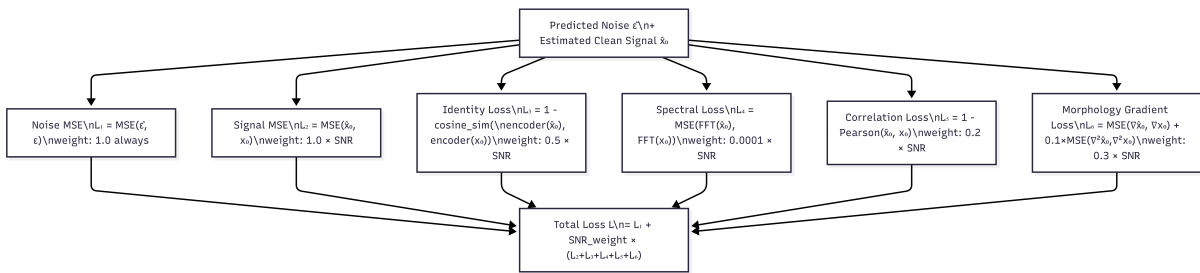
4.4 Output Head



Full DiT-ECG-B specs:

Parameter	Value
Transformer blocks	24
Model dimension <code>d_model</code>	768
Attention heads	12
FFN expansion	4× (3072)
Patch size	10 (250 tokens)
Total parameters	265M
Precision	BF16 mixed

5. Loss Function — Multi-Component Training Signal



SNR Weighting: Auxiliary losses ($L_i - L_m$) are weighted by α_{t} (signal-to-noise ratio at timestep t). At high noise levels (large t), $\alpha_{\text{t}} \approx 0$ so auxiliary losses are suppressed — only the denoising loss L_d matters. At low noise (small t), $\alpha_{\text{t}} \approx 1$ so all losses contribute equally. This ensures losses are applied only when the reconstructed signal $\hat{x}_{\text{t}}$ is reliable.

Run 4 Bug Found: Spectral loss was producing values ~125 while all others were ~0.1. The weight was changed from 0.1 → 0.0001 to fix this.

6. Forward & Reverse Diffusion

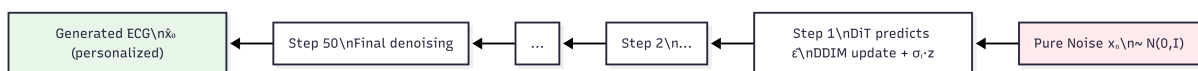
Forward Process (Training)



Formula: $x_t = \sqrt{\alpha_t} \cdot x_0 + \sqrt{(1-\alpha_t)} \cdot \epsilon$ where $\epsilon \sim N(0, I)$

Uses **cosine noise schedule** (Nichol & Dhariwal, 2021) — smoother than linear, better for ECG morphology at fine scales.

Reverse Process (Inference — DDIM with eta)



DDIM with $\eta=0.75$: Standard DDIM uses $\eta=0$ (deterministic). We use $\eta=0.75$ to re-inject stochastic noise at each step:

$$\sigma_t = \eta \times \sqrt{((1-\alpha_{t+1}))/((1-\alpha_t)) \times ((1-\alpha_t)/\alpha_{t+1}))}$$

This increases **HR diversity** by preventing the model from always taking the same deterministic path.

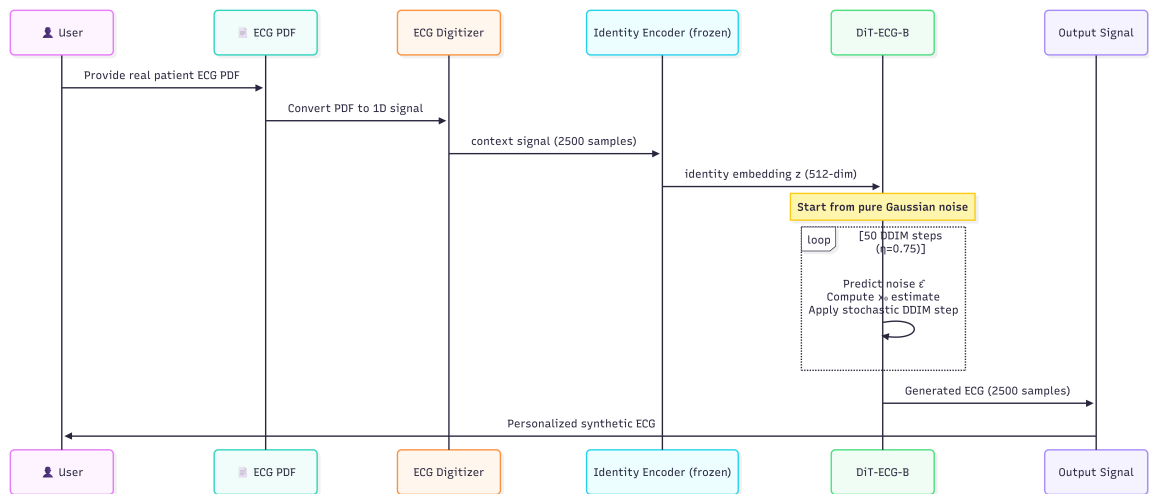
7. Identity Encoder: ContrastiveFeatureExtractor



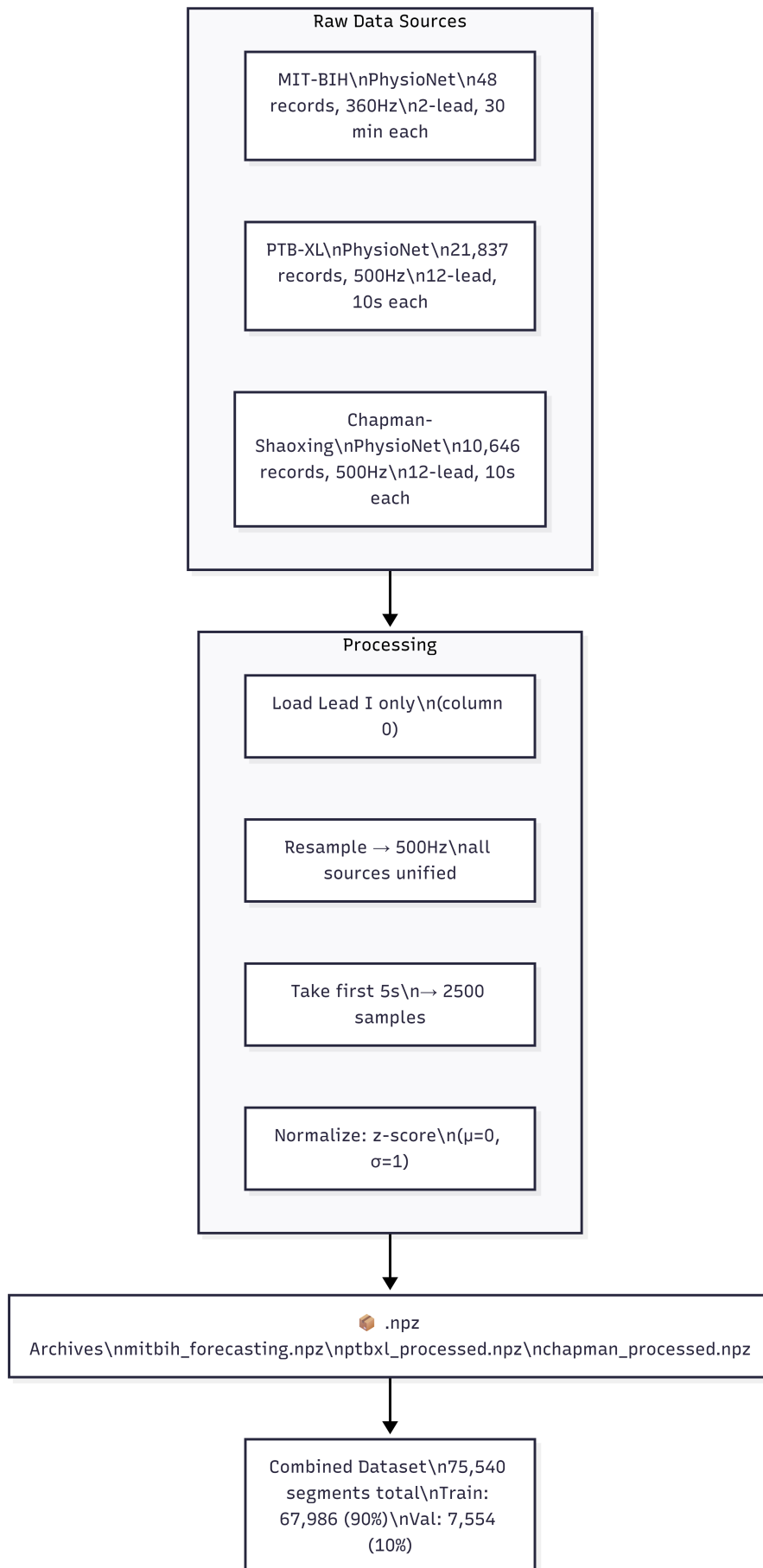
Training: InfoNCE contrastive loss. Two random crops from the **same patient** are pulled together in embedding space; crops from **different patients** are pushed apart.

At inference: Encoder is **frozen** — its weights do not change during DiT training. This ensures the identity space is stable and meaningful.

8. Inference Pipeline



9. Data Flow & Preprocessing



10. Training Configuration

Setting	Value	Why
Batch size	16 (micro)	GPU memory constraint (A100 80GB shared)
Gradient accumulation	16 steps	Effective batch = 256
Learning rate	1e-4	AdamW, cosine decay
Warmup steps	5,000	Stable LR ramp-up
Max epochs	500	Early stopping at patience=30
EMA decay	0.9999	Smoothed weights for inference
CFG dropout	10%	Classifier-Free Guidance training
Diffusion timesteps	1,000	Standard DDPM/DDIM
DDIM steps (inference)	50	Speed/quality tradeoff
DDIM eta	0.75	HR diversity (new in Run 4)
Precision	BF16	~2× memory saving, minimal quality loss

11. Evaluation Metrics

Metric	What It Measures	Run 1	Run 2	Run 4
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FFD ↓	Overall signal quality (like FID for images)	1032	42.0	21.7
MMD ↓	Distribution mismatch (real vs generated)	0.661	0.383	0.432
HR MAE ↓	Mean absolute heart rate error (bpm)	17.2	9.3	31.4*
HR Std	Diversity of generated heart rates	—	21.1	20.5
Real HR Std	Target diversity	—	—	47.4
ReID Top-1 ↑	Patient identity preserved (correct ID)	11.8%	5.9%	11.8%
ReID Top-5 ↑	Patient in top-5 predictions	35.3%	23.5%	35.3%

*HR MAE worsened due to spectral loss imbalance bug (now fixed for Run 5)

12. Full System File Map

```
CardioEquation/  
  src/  
    models/  
      dit_ecg.py ← Main generator (265M params)  
      feature_extractor_pt.py ← Identity encoder (9.4M params)
```

```

■   ■■■ training/
■   ■   ■■■ train_contrastive.py    ← Stage 0: Identity pre-training
■   ■   ■■■ train_dit.py           ← Stage 1: Diffusion training
■   ■   ■■■ losses_v2.py           ← 6-component loss functions
■   ■   ■■■ noise_scheduler.py      ← Cosine schedule + DDIM sampler
■   ■   ■■■ ema.py                 ← Exponential Moving Average
■   ■■■ inference/
■   ■   ■■■ pipeline_v2.py          ← End-to-end generation pipeline
■   ■■■ evaluation/
■       ■■■ clinical_validation.py  ← Hospital ECG evaluation
■■■ data/                          ← Downloaded datasets (.npz)
■■■ checkpoints/                   ← Saved model weights
■■■ docs/                          ← This document + future techniques
■■■ run_training.sh                ← One-command full pipeline

```

13. Key Design Decisions & Rationale

Decision	Alternative Considered	Why We Chose This
DiT backbone	U-Net (standard DDPM)	DiT scales better: bigger model = better quality. State-of-the-art for image generation (DALL-E 3, Stable Diffusion 3)
Contrastive identity encoder	Siamese network, AE	InfoNCE produces better-separated patient embeddings. Used in ECGTwin (2025)
Cosine noise schedule	Linear	Cosine is smoother at low noise levels — better for ECG fine structure (P-waves)
DDIM sampling	Full DDPM	50 DDIM steps ≈ quality of 1000 DDPM steps. 20× faster inference

$\eta=0.75$ stochastic DDIM	$\eta=0$ deterministic	Deterministic DDIM collapses HR diversity. Stochasticity forces varied outputs
SNR-weighted auxiliary losses	Fixed weights	Prevents auxiliary instability at high noise timesteps. Proven in iDDPM (Nichol 2021)
Frozen encoder during DiT training	Joint fine-tuning	Fine-tuning corrupts carefully learned identity representations

14. References

- **DiT**: Peebles & Xie (2022). *Scalable Diffusion Models with Transformers*. ICCV 2023.
- **DDIM**: Song et al. (2020). *Denoising Diffusion Implicit Models*. ICLR 2021.
- **iDDPM**: Nichol & Dhariwal (2021). *Improved DDPM*. ICML 2021.
- **ECGTwin**: (2025). *Personalized ECG generation using controllable diffusion*. arXiv:2508.
- **DiffuSETS**: Lai et al. (2025). *12-lead ECG generation conditioned on clinical text*. Patterns.
- **InfoNCE**: Oord et al. (2018). *Representation Learning with Contrastive Predictive Coding*.
- **Cosine schedule**: Nichol & Dhariwal (2021). *Improved DDPM*.
- **CFG**: Ho & Salimans (2022). *Classifier-Free Diffusion Guidance*.